



Original article

Detection of metabolic signatures implicated in the progression from hepatitis to cirrhosis to hepatocellular carcinoma

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ABSTRACT

Background and aims: Hepatocellular carcinoma (HCC) is the third leading cause of cancer-related deaths worldwide. The mechanisms driving the transition from hepatitis to cirrhosis, and eventually, to HCC are unclear. This study aimed to clarify the metabolic changes that underly the progression of HCC and identify potential prognostic and therapeutic biomarkers.

Methods: This prospective study collected serum samples from patients with chronic hepatitis, cirrhosis, or HCC, hospitalized at the Fifth Medical Center of the PLA General Hospital, from December 2022 to December 2023. The samples were analyzed using non-targeted, ultra-high-performance liquid chromatography and mass spectrometry. Partial least squares-discriminant analysis modeling and *t*-tests were used to identify key differentially expressed metabolites associated with the progression from hepatitis to cirrhosis to HCC. Pathway enrichment analysis was conducted to determine the key metabolic pathways involved, while machine learning models were applied to identify the metabolite signatures.

Results: We identified 153 differentially expressed metabolites in the progression from hepatitis to cirrhosis to HCC, many of which were involved in ammonia cycling or the metabolism of methylhistidine, alanine, arginine, proline, or betaine. We also identified L-histidine and adenosine as the metabolites that demonstrated significant sensitivity and specificity for distinguishing among the hepatitis, cirrhosis, and HCC stages.

Conclusions: Our study comprehensively characterized the metabolic profiles of the different stages of the hepatitis-cirrhosis-HCC transition. We showed that serum metabolite detection is a viable diagnostic tool for identifying and monitoring high-risk individuals, which could potentially be used to halt the development of HCC.

1. Introduction

Primary liver cancer (PLC) is a major global health challenge and is the third leading cause of cancer-related deaths worldwide.¹ Hepatocellular carcinoma (HCC) represents the predominant histologic subtype of PLC, comprising over 80% of all PLC cases.¹ Chronic infection with the hepatitis B virus (HBV) is the primary and most potent risk factor for HCC, with at least 60% of HCC cases attributable to HBV infection.² The progression from chronic HBV infection to HCC is a multistep process. Patients typically develop hepatitis, then cirrhosis, and ultimately, HCC.³ While the progression from hepatitis to HCC has become a major research focus, the underlying mechanisms driving this transition remain poorly

understood. Clarifying the pathogenesis of HCC in patients with hepatitis and cirrhosis will help facilitate the early diagnosis HCC, as well as identify novel therapeutic targets.

Metabolic reprogramming, during which cancer cells undergo metabolic alterations to sustain tumorigenesis and promote cell proliferation, is a hallmark of cancer.⁴ During carcinogenesis, cellular metabolism is altered in a way that enables the cancer cell to adapt to a hypoxic environment and evade proapoptotic signals.⁵ Increasing evidence suggests that specific metabolites can promote cancer by modulating various signaling pathways⁶; however, their exact mechanisms of action remain unclear. Therefore, studying aspects of metabolic reprogramming which characterize cancer progression from hepatitis to HCC may help uncover

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novel metabolic markers. This approach could also facilitate the identification of potential biomarkers of early cirrhosis in patients with chronic hepatitis, as well as predictive biomarkers of progression from cirrhosis to HCC.

Metabolomics, which aims to comprehensively characterize the entire metabolome of biological samples, is a promising new analytical discipline in the field of precision diagnostics.⁷ The advancement of high throughput metabolomic technologies has enabled the generation of vast quantities of molecular data. These large datasets will aid in the elucidation of biochemical perturbations in various cancers, facilitating the discovery of potential therapeutic targets and diagnostic biomarkers.⁸ Metabolites that are readily accessible in peripheral circulation can offer rapid insights into an individual's hepatic metabolic status. Several metabolomic studies have explored the discriminatory metabolic profiles of patients with chronic hepatitis, cirrhosis, or HCC.^{9,10} However, a systematic investigation of the metabolic profiles that characterize each stage of disease, and their interactions throughout the hepatitis-cirrhosis-HCC transition, has not been performed to date. Therefore, the aim of the present study was to perform serum metabolomics profiling of patients with chronic hepatitis, cirrhosis, or HCC, to elucidate the biological mechanisms involved in HCC progression. By identifying metabolic trends, we aimed to discover potential metabolic biomarkers that could be used to guide treatment decisions, assess treatment efficacy, and establish objective criteria for diagnosing and treating patients progressing from hepatitis to cirrhosis to HCC.

2. Materials and methods

2.1. Study design and participants

A total of 379 participants were enrolled in the study from The Fifth Medical Center of PLA General Hospital and the Dongzhimen Hospital (Beijing University of Chinese Medicine) between December 2022 and December 2023. The cohort comprised 52 healthy controls (HCs), 81 patients with chronic (CHB), 180 patients with hepatitis-B-associated cirrhosis (HBC), and 66 patients with HCC. This study was approved by the Ethics Committee of Beijing University of Chinese Medicine (approval no. 2022BZYLL1206) and adhered to the ethical guidelines of the 1975 Declaration of Helsinki and its amendments. Written informed consent was obtained from each participant.

The inclusion criteria for the study were as follows: participants aged 18–75 years, with HBV infection and defined CHB, liver cirrhosis, or HCC. The exclusion criteria were: current or historical conditions of other liver diseases such as hepatitis C, nonalcoholic/alcoholic liver disease, or autoimmune liver disease; concurrent malignancies; severe cardiovascular, neurological, pulmonary, renal, hematologic, or psychiatric disorders; decompensated cirrhosis; patients with definite HCC receiving anti-cancer therapy; and pregnant or lactating women.

HCC diagnosis was confirmed using ultrasound, computed tomography, or magnetic resonance imaging.¹¹ Liver cirrhosis was defined as

imaging findings indicative of liver cirrhosis and/or portal hypertension, with exclusion of non-cirrhotic causes of portal hypertension.¹² Patients with liver cirrhosis were classified as having decompensated cirrhosis if they experienced severe complications such as esophageal or gastric variceal bleeding, ascites, or hepatic encephalopathy.¹² CHB was defined by laboratory findings of serum hepatitis B surface antigen positivity and HBV deoxyribonucleic acid (DNA) positivity, along with persistent or recurrent raised levels of alanine aminotransferase (ALT) or aspartate aminotransferase (AST).¹²

Demographic characteristics, including sex, age, body mass index (BMI), were recorded for each study participant (Table 1). Blood indicators, such as the serum levels of HBV DNA, ALT, AST, total bilirubin (TB), and albumin (ALB), as well as platelet (PLT) counts, were also documented. The sex and age distributions among the four groups were well matched.

2.2. Sample collection

All the samples were collected from patients who had fasted for more than 8 h and were obtained fresh on the morning of post-admission day 2. Blood was collected in coagulation tubes. After collection, the tubes were gently inverted and centrifuged at 3000 rpm for 10 min at room temperature. The supernatant (serum) was then transferred to 1.5 mL cryovials, labeled with the sample information, and stored at –80 °C until further analysis.

2.3. Metabolite extraction

Next, 100 µL of the serum sample was placed in an Eppendorf tube and diluted thoroughly with pre-chilled 80% methanol. The sample was then incubated on ice for 5 min, before being centrifuged at 15,000×g, 4 °C for 20 min. A portion of the supernatant was then diluted with liquid chromatography-mass spectrometry (LC-MS)-grade water so that the final concentration of methanol was reduced to 53%. The sample was then transferred to a new Eppendorf tube and centrifuged again at 15,000×g, 4 °C for 20 min. Finally, the supernatant was injected into the Ultra High Performance LC-MS (UHPLC-MS/MS) system for analysis.

2.4. UHPLC-MS/MS analysis

The UHPLC-MS/MS analysis was performed using the Vanquish UHPLC system and the Orbitrap Q Exactive™ HF-X mass spectrometer (both from Thermo Fisher Scientific, USA). Samples were injected onto a Hypersil Gold column (100 × 2.1 mm, 1.9 µm) and separated on a linear gradient over 17 min at a flow rate of 0.2 mL/min. The raw data files generated from the UHPLC-MS/MS analysis were processed using Compound Discoverer 3.1 (CD3.1, Thermo Fisher Scientific). The analysis involved peak alignment, peak selection, and the quantification of each metabolite in the sample.

Table 1
Clinical characteristics of the study subjects.

Characteristics	HC group (n = 52)	CHB group (n = 81)	HBC group (n = 180)	HCC group (n = 66)
Male sex, n (%)	38 (73.1)	60 (74.1)	132 (73.3)	50 (75.8)
Female sex, n (%)	14 (26.9)	21 (25.9)	48 (26.7)	16 (24.2)
Age, median (IQR)	54.0 (47.0,66.0)	54.0 (49.5,59.5)	56.0 (48.3,65.0)	58.5 (51.0,65.3)
BMI, kg/m ² , median (IQR)	24.5 (21.7,26.8)	23.4 (20.8,25.4)	24.2 (22.5,26.3)	24.4 (23.3,26.3)
Positive HBVDNA, n (%)	0	17 (21.0)	23 (12.8)	9 (13.6)
ALT, U/L, median (IQR)	18.0 (13.0,31.0)	19.0 (14.0,26.0)	23.0 (16.0,34.0)	20.5 (14.0,31.0)
AST, U/L, median (IQR)	22.0 (19.0,37.0)	26.0 (20.5,36.0)	29.0 (21.3,42.8)	25.0 (20.0,38.3)
TB, µmol/L, median (IQR)	13.4 (10.3,16.5)	18.2 (12.8,16.6)	16.4 (12.0,26.0)	19.3 (11.7,32.1)
ALB, g/L, median (IQR)	40.0 (38.0,42.0)	37.0 (33.0,40.5)	37.0 (32.0,41.0)	38.0 (34.0,40.3)
PLT, ×10 ⁹ /L, median (IQR)	225.0 (193.5,260.0)	173.0 (118.5,221.5)	95.5 (56.0,137.0)	121.5 (69.0,145.8)

Abbreviations: IQR, interquartile range; BMI, body mass index; HBVDNA, hepatitis B virus deoxyribonucleic acid; ALT, alanine aminotransferase; AST, aspartate aminotransferase; TB, total bilirubin; ALB, albumin; PLT, platelet.

2.5. Statistical analysis

Statistical analyses were conducted using R (version 4.3.2), Python (version 3.11.0), and CentOS (version 7.0). For non-normally distributed data, normalization was performed using area normalization methods. Metabolites were annotated using the KEGG (<https://www.genome.jp/kegg/pathway.html>) and HMDB (<https://hmdb.ca/metabolites>) databases. A 40% interquartile range (IQR) filter was used to remove low-variance features. The samples were then normalized by variance-stabilized transformation (base 10). Finally, auto-scaling was applied by mean-centering and dividing by the standard deviation. Statistical significance was calculated using *t*-tests. Metabolites with a VIP >1, a *p*-value <0.05, and a $\log_2$ fold change (FC) ≥ 1.25 were considered as significantly different. Volcano plots were generated using the ggplot2 package in R, based on $\log_2$ (FC) and $\log_{10}$ (*p*-value), to identify the metabolites of interest. The random forest model was configured with 500 trees and seven predictors (features). Randomness was enabled, allowing the model to select random subsets of predictors at each split to enhance model robustness and generalizability. Receiver operating characteristic (ROC) curve analysis was conducted to assess the diagnostic performance of key metabolites in distinguishing between disease stages. Area under the ROC curve (AUC) values were calculated as a measure of sensitivity and specificity. A functional analysis of the metabolites of interest and their metabolic pathways was performed using the HMDB database. Pathway enrichment was evaluated based on the criterion $x/n > y/N$, where significantly enriched pathways had a *p*-value <0.05.

3. Results

3.1. Comparing the serum metabolomic profiles of HBV-infected patients and HCs

The UPLC-MS analysis detected a total of 20,287 non-targeted peaks. After rigorous quality control and identification, 1590 serum metabolites were annotated. Principal component analysis (PCA) revealed significant differences in the serum metabolite composition of HBV-infected patients (the CHB, HBC, and HCC groups) and HCs (Fig. 1A). Further validation with partial least-squares discriminant analysis (PLS-DA) models confirmed these differences (Fig. 1B). Using the VIP >1.0, $\log_2$ (FC) ≥ 1.25 , and *p* < 0.05 criteria, 970 (304 upregulated and 666 downregulated) differentially expressed metabolites were identified between the HBV-infected patients and the HCs (Fig. 1C and Supplementary Table S1).

3.2. Comparing the serum metabolomic profiles of the CHB, HBC, and HCC groups

Next, we conducted a PCA analysis of the 970 differentially expressed serum metabolites between HBV-infected patients and HCs. This analysis also revealed significant differences in serum metabolomic profiles among the CHB, HBC, and HCC groups (Fig. 2A). The PLS-DA model further confirmed these significant differences (Fig. 2B). Using the VIP >1.0, $\log_2$ (FC) ≥ 1.25 , and *p* < 0.05 criteria, we identified 522 differentially expressed metabolites between the CHB and HBC groups

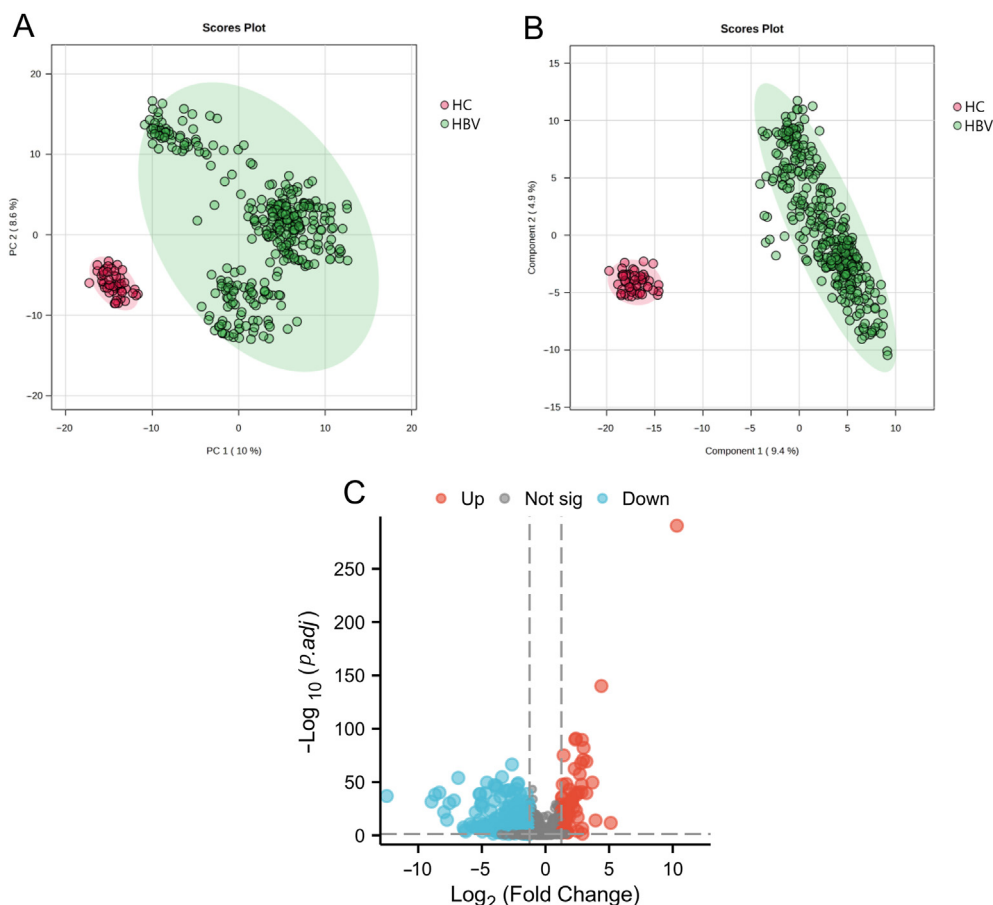


Fig. 1. Serum metabolomics comparison between hepatitis-B-virus-infected patients and healthy controls. (A) PCA score scatter plots, (B) PLS-DA score scatter plots, and (C) volcano plot of differentially expressed metabolites between HBV-infected patients and healthy control subjects. Abbreviations: HC, healthy control; HBV, hepatitis B virus; PCA, principal component analysis; PLS-DA, partial least-squares discriminant analysis.

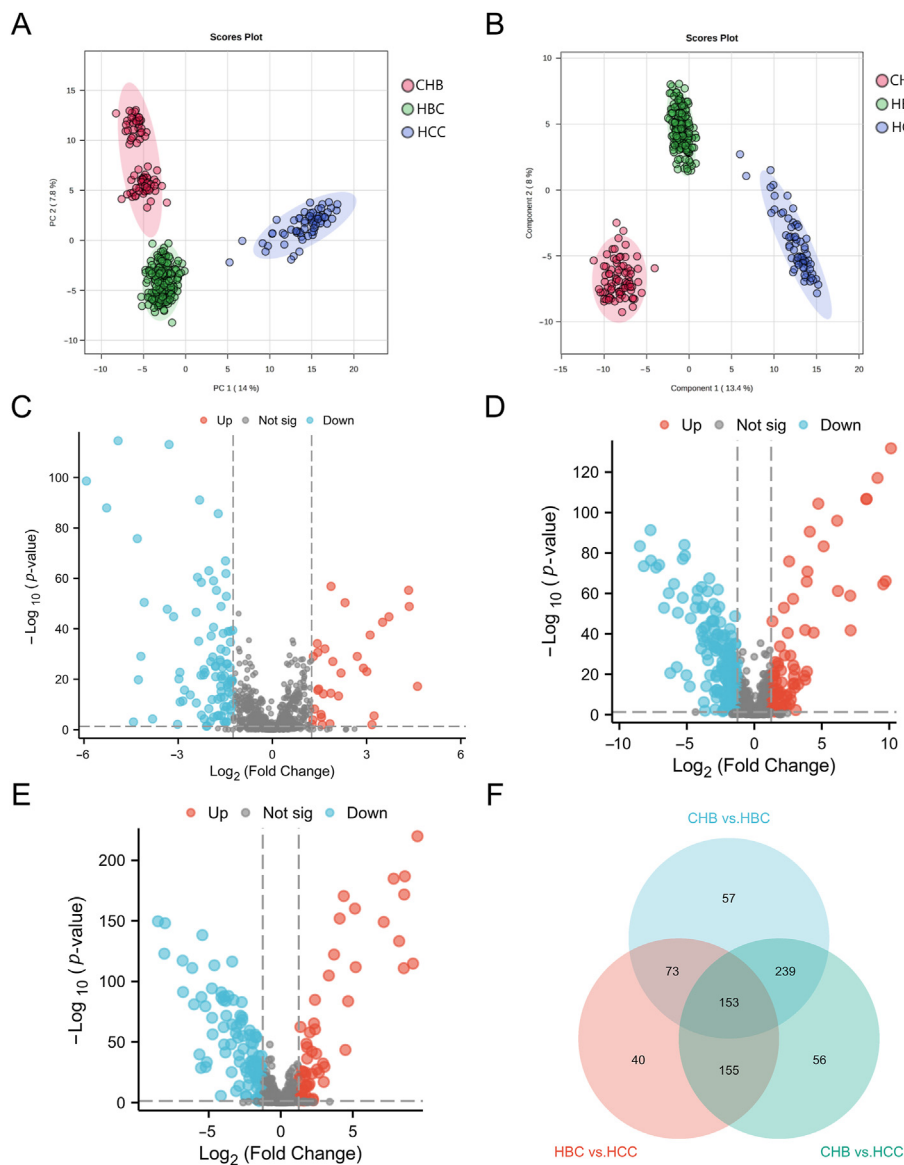


Fig. 2. Assessment of serum metabolites during the progression from CHB to HBC to HCC. (A) PCA scatter plots and (B) PLS-DA scatter plots for the CHB, HBC, and HCC groups. Volcano plot of differentially expressed metabolites between the (C) CHB and HBC groups, (D) the CHB and HCC groups, and (E) the HBC and HCC groups; (F) Venn diagram of the differentially expressed metabolites shared among the CHB, HBC, and HCC groups. Abbreviations: CHB, chronic hepatitis B; HBC, hepatitis B cirrhosis; HCC, hepatocellular carcinoma; PCA, principal component analysis; PLS-DA, partial least-squares discriminant analysis.

(Fig. 2C), 603 between the CHB and HCC groups (Fig. 2D), and 421 between the HBC and HCC groups (Fig. 2E). Notably, 153 differentially expressed metabolites were shared by all three groups of HBV-infected patients (Fig. 2F and Supplementary Table S2).

3.3. Identifying key metabolic changes that occur in the progression from hepatitis to cirrhosis to HCC

We then used the Random Forest (RF) machine learning algorithm to determine the roles of the 153 shared differentially expressed metabolites in the progression from hepatitis to cirrhosis to HCC. In the RF model, metabolite importance is assessed by the mean decrease accuracy (MDA) value, which is a measure of how reliably a metabolite can be used to discriminate between groups. The higher the MDA value, the greater the ability of a metabolite to distinguish between groups. Metabolites with an MDA value > 0.01 were selected for the RF analysis of the CHB, HBC, and HCC groups. The results showed that L-histidine (MDA = 0.014326) and adenosine (MDA = 0.013314) were key metabolites in the progression from hepatitis to cirrhosis to HCC. While adenosine levels increased with each HCC progression stage, those of L-histidine decreased across the stages (Fig. 3A). The results of the ROC analysis confirmed that both metabolites had significant sensitivity and

specificity in distinguishing among the three stages of HCC development. L-histidine had an AUC of 0.999 for CHB vs. HBC and 0.990 for HBC vs. HCC (Fig. 3B). Adenosine had an AUC of 1.000 for CHB vs. HBC and 0.944 for HBC vs. HCC (Fig. 3C).

3.4. Identifying the key metabolic pathways implicated in the progression from hepatitis to cirrhosis to HCC

To further understand the metabolic changes associated with the progression from hepatitis to cirrhosis to HCC, we performed enrichment analyses of differentially expressed metabolites between the HCC and the CHB/HBC groups (Fig. 4A), between the CHB and HBC groups (Fig. 4B), and between the HBC and HCC groups (Fig. 4C). We found significant alterations in the metabolic pathways, including ammonia cycling, methylhistidine metabolism, alanine metabolism, arginine and proline metabolism, and betaine metabolism, across all comparisons (HCC vs. CHB/HBC, CHB vs. HBC, and HBC vs. HCC). These metabolic pathway changes were observed throughout the progression from hepatitis to cirrhosis to HCC. This suggests that the metabolic reprogramming contributing to HCC development may occur as early as during the CHB stage. Thus, the dysregulation of ammonia cycling, methylhistidine metabolism, alanine metabolism, arginine and proline metabolism, and

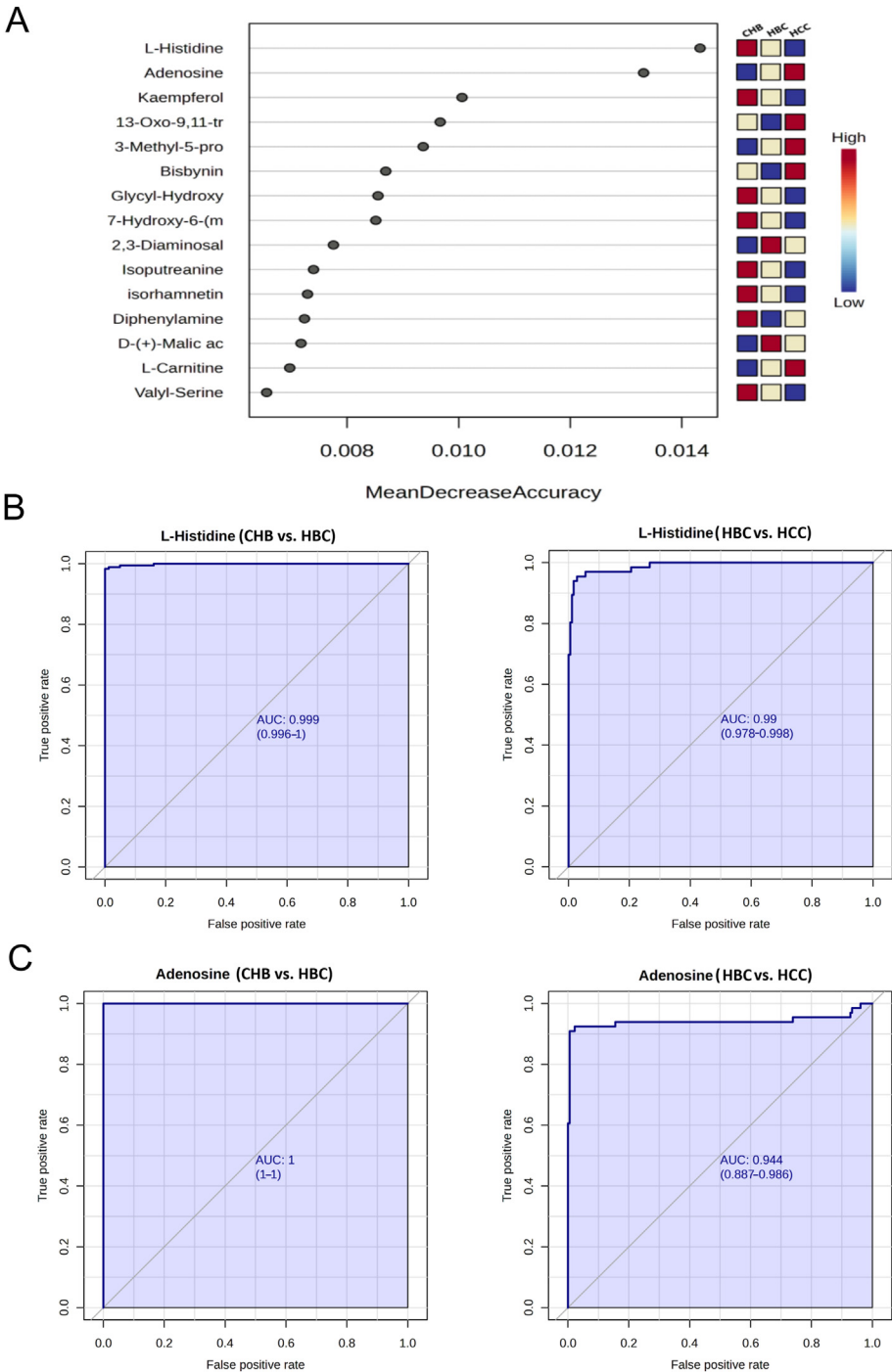


Fig. 3. Key metabolites involved in the progression from CHB to HBC to HCC. (A) RF analysis of differentially expressed metabolites during the progression from CHB to HBC to HCC. AUC curves were generated for (B) L-histidine and (C) adenosine, to assess their specificity and sensitivity in differentiating among the three stages of HCC development. Abbreviations: CHB, chronic hepatitis B; HBC, hepatitis B cirrhosis; HCC, hepatocellular carcinoma; RF, random forest; AUC, area under the receiver operating characteristic curve.

betaine metabolism may contribute to the progression from hepatitis to cirrhosis to HCC.

4. Discussion

Approximately 80%–90% of HCC cases occur in the context of cirrhosis, underscoring the significant roles of hepatitis and cirrhosis in the precancerous liver environment.¹³ However, few studies have used comprehensive metabolic profiling during the progression from hepatitis to cirrhosis to HCC to identify the key metabolites and pathways implicated in this process. Our study was the first to characterize the dynamic changes in metabolite levels and distribution that occur throughout the transition from hepatitis to cirrhosis to HCC. By revealing that metabolic

reprogramming is a key driver of HCC progression, these findings will help further our understanding of HCC pathogenesis.

In this study, we used a non-targeted metabolomics analysis method to successfully identify 153 key metabolites involved in HCC progression. Moreover, we showed that these metabolites were enriched in pathways implicated in the cycling of ammonia and the metabolism of methyl-histidine, alanine, arginine, proline, and betaine. Changes in these metabolic pathways were evident throughout the hepatitis-cirrhosis-HCC transition. These observations suggest that metabolic reprogramming that occurs in patients with chronic hepatitis could potentially be a driving factor in the progression of HCC.

Our machine learning analysis identified L-histidine and adenosine as two metabolites which could accurately differentiate among the three

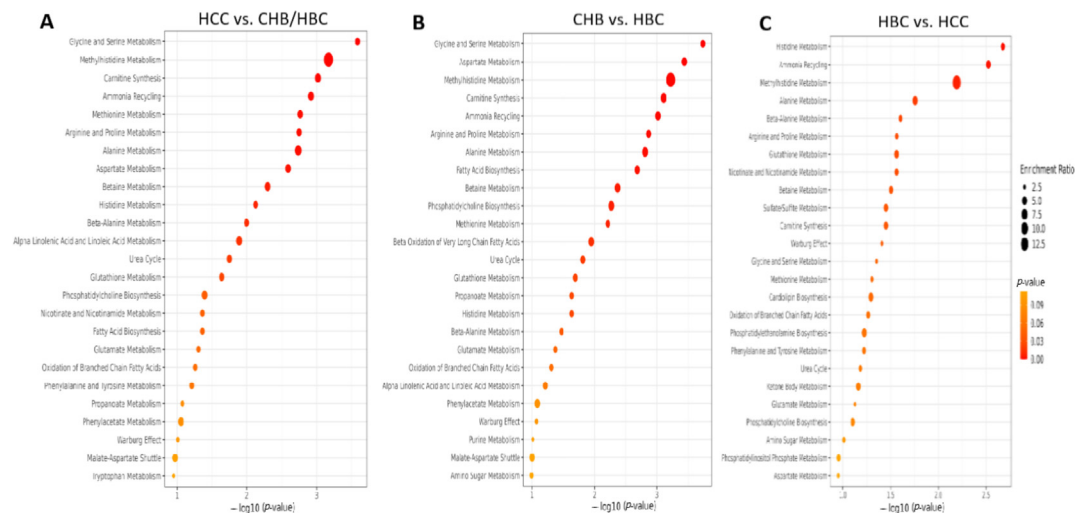


Fig. 4. Pathway enrichment analysis of differentially expressed metabolites identified at each stage of HCC progression. The metabolic pathways altered in (A) the HCC group versus the CHB/HBC groups, (B) the CHB group versus the HBC group, and (C) the HBC group versus the HCC group. Abbreviations: CHB, chronic hepatitis B; HBC, hepatitis B cirrhosis; HCC, hepatocellular carcinoma; RF, random forest; AUC, area under the receiver operating characteristic curve.

stages of HCC development. L-histidine and adenosine demonstrated high sensitivity and specificity for distinguishing between patients with chronic hepatitis and those with cirrhosis, as well as between patients with cirrhosis and those with HCC. Thus, L-histidine and adenosine represent potentially important diagnostic biomarkers for evaluating the progression from hepatitis to cirrhosis to HCC.

L-histidine is an essential amino acid that plays a crucial role in various biological functions within the human body. Previous research has established a close relationship between histidine metabolism and the pathophysiological progression of liver disease.¹⁴ The degradation of histidine in the body involves histidine ammonia-lyase, an enzyme which is implicated in the first step of histidine metabolism, and which is found exclusively in the liver and the epidermis. This enzyme catalyzes the removal of the α -amino group from L-histidine, which results in the formation of ammonia and urocanic acid.¹⁵ Additionally, histidine can be decarboxylated into histamine by L-histidine decarboxylase, an enzyme present in various tissues, including the colon, liver, lungs, muscles, and gastric mucosa.¹⁶ Histidine can also cross the blood-brain barrier, before being converted into histamine by L-histidine decarboxylase in the hypothalamic tuberomammillary nucleus. Depending on the metabolic pathway utilized, histidine can be converted into either carnosine or histamine, both of which have important roles in the body.¹⁶ Moreover, histamine can scavenge reactive oxygen species (ROS) and inactivate the highly reactive α,β -unsaturated aldehydes which are formed during oxidative stress.¹⁷ L-histidine can also increase antioxidant levels, eliminate free radicals in the liver, restore mitochondrial function, and alleviate liver inflammation by reducing the concentrations of the enzymes ALT and AST.¹⁸ Hinrichs et al. found that increasing L-histidine levels by knocking out L-histidine decarboxylase inhibited oxidative stress and reduced liver inflammation.¹⁹ Moreover, in accordance with our findings, several studies have observed a decrease in L-histidine levels in patients with cirrhosis and liver cancer.^{20, 21} Park et al. discovered that administering histidine to patients with HCC reduced their expression of various tumor markers associated with glycolysis (GLUT1 and HK2), inflammation (STAT3), angiogenesis (VEGFB and VEGFC), and stem cells (CD133). Crucially, histidine supplementation also increased patient sensitivity to chemotherapeutic agents such as sorafenib, as well as their long-term survival.²² In summary, L-histidine can reduce both the liver inflammation and the risk of progression from hepatitis to cirrhosis to HCC. Our findings suggest that a progressive decline in L-histidine levels may be the key metabolic change driving progression from hepatitis to cirrhosis to HCC. Consequently, histidine supplementation might delay or block this progression.

Adenosine is a nucleoside widely present in most cell types. Upon its release by metabolically active or stressed cells, adenosine binds to specific adenosine receptors on the cell surface.²³ Almost all cells express specific adenosine receptors, and as an important physiological regulator, adenosine exhibits cardioprotective, neuroprotective, chemoprotective, and immunomodulatory activities.²⁴ Additionally, adenosine possesses immunosuppressive functions.²⁵ Infection triggers the infiltration of immune cells into tissues to eradicate the invading pathogen. Later in the immune response, immunosuppressive cells assist in reducing inflammation and promoting healing to restore tissue integrity.²⁶ Immune cells produce immunosuppressive mediators and cytokines to repair tissue damage and stimulate the proliferation of previously quiescent cells near the site of infection.²⁷ One such immunosuppressive mediator is adenosine. Adenosine can be released by cells or generated extracellularly through the sequential hydrolysis of ATP by the ecto-nucleotide triphosphate diphosphohydrolases CD39 and CD73, or through pore-mediated release.²⁸ Although CD39 and CD73 are expressed on various types of immune cells (e.g., T cells, dendritic cells [DCs], B cells, and neutrophils), they are especially abundant on immunosuppressive cells, including regulatory T cells (Tregs), immature DCs, and inhibitory B cells. These cells can dephosphorylate extracellular ATP into ADP and AMP, which are ultimately degraded into adenosine.²⁹ Adenosine exerts its anti-inflammatory function on a variety of immune cell types. It can inhibit the production of macrophage-colony-stimulating factor and suppress the activation and phagocytic function of M1 macrophages. In doing so, adenosine promotes the polarization of pro-inflammatory M1 macrophages, which are initially recruited to inflamed tissues, into the anti-inflammatory M2 macrophage subtype.³⁰ Furthermore, adenosine has been reported to induce the proliferation and activation of tumor-associated macrophages in HCC, thereby promoting HCC progression.³¹ Adenosine A2 receptors can indirectly suppress the anti-tumor cytotoxicity of T and NK cells by reducing their secretion of interleukin (IL)-12 while increasing their production of IL-10.³² Adenosine can also significantly enhance the recruitment of plasmacytoid DCs to HCC tumors via its action on adenosine A1 receptors. This interaction facilitates the immune evasion of HCC tumors by promoting the induction of Tregs while suppressing CD8⁺ T cell proliferation and cytotoxicity.³³ As a major inducer of Treg-mediated immunosuppression, adenosine is crucial for downregulating inflammatory responses and preventing excessive immune reactions. Studies have confirmed that the absence of CD73, which catalyzes adenosine production, impairs the immunosuppressive function of Tregs, which promotes HCC tumor growth and metastasis.³⁴

Additionally, adenosine has been shown to enhance collagen production by fibroblasts, contributing to the progression from chronic hepatitis to liver fibrosis and cirrhosis.³⁵ In summary, adenosine is closely linked to the progression from hepatitis to cirrhosis to HCC. During this process, adenosine may initially prevent the immune system from clearing pathogens or harmful substances, thus facilitating the transition from acute to chronic inflammation. Moreover, some tumors express the key adenosine-producing enzymes CD39 and CD73, or acquire additional adenosine reserves via other means, to evade immune detection.²⁸ The combination of persistent chronic inflammation and high adenosine levels creates a tissue environment that favors HCC development. Various metabolic modulation methods and means of targeting adenosine receptors have been developed to regulate adenosine levels.³⁶ Targeting the adenosine/adenosine receptor pathway may prove to be an effective therapeutic strategy for inhibiting hepatitis-cirrhosis-HCC progression.

5. Conclusions

The metabolites found in the peripheral circulation are readily accessible. As such, they can provide valuable insights into the metabolic changes that occur during the progression from hepatitis to cirrhosis to HCC. In the present study, we performed serum metabolomics analysis to comprehensively characterize the metabolic changes contributing to HCC development. This led to the identification of five key altered metabolic pathways (i.e., ammonia cycle, methylhistidine metabolism, alanine metabolism, arginine and proline metabolism, and betaine metabolism) along with two key metabolites (L-histidine and adenosine). Our findings elucidate the metabolic mechanisms underlying the hepatitis-cirrhosis-HCC transition and highlight the potential utility of the metabolites identified in the risk stratification and early diagnosis of HCC patients.

We acknowledge that our study has several limitations. First, similar to other metabolomics studies, the metabolites were measured at a single timepoint, which may not fully capture the metabolic changes that occur during long-term disease. Second, although we established correlations between metabolic changes and disease, we could not prove causation.

Future research will focus on determining how to effectively integrate metabolite detection into routine clinical diagnostic procedures. We aim to develop standardized testing methods to ensure the reproducibility and reliability of this strategy in the laboratory setting. Clinical validation studies will be conducted to assess the accuracy and effectiveness of these metabolites in specific diseases, helping to establish their clinical significance and provide guidance for healthcare professionals. Additionally, we recognize the importance of external validation in improving the performance of our machine learning models. In future studies, we will implement cross-validation methods to ensure the reliability of the results. This approach will enhance our confidence in the predictive capabilities of these models and their applicability in clinical practice. In addition, we plan to validate the clinical effectiveness of the metabolites identified in this study by conducting large-scale prospective cohort studies, which will gather data from diverse populations. Collaboration with multiple clinical centers will further allow us to examine the role of specific metabolites across different disease stages and their correlation with clinical outcomes. Long-term follow-up studies will help us understand how fluctuations in metabolite levels impact patient prognosis, thereby confirming their potential as biomarkers. These efforts will lay the groundwork for incorporating metabolite detection into standard clinical practice and will drive further research to validate their clinical utility, ultimately providing more precise diagnostic and therapeutic options for patients.

CRediT authorship contribution statement

Simiao Yu: Writing – original draft. **Sici Wang:** Visualization, Methodology. **Jiahui Li:** Investigation, Formal analysis. **Haocheng Zheng:** Visualization, Conceptualization. **Ping Li:** Visualization. **Wenya Rong:** Investigation. **Jing Jing:** Investigation. **Tingting He:**

Investigation. **Yongqiang Sun:** Investigation. **Liping Wang:** Investigation. **Zhenyu Zhu:** Writing – review & editing, Supervision. **Xia Ding:** Writing – review & editing, Supervision, Conceptualization. **Ruilin Wang:** Writing – review & editing, Supervision, Conceptualization.

Informed consent

All the enrolled patients provided written informed consent.

Ethics statement

This study was approved by the Ethics Committee of Beijing University of Chinese Medicine (approval no. 2022BZYLL1206). All procedures were performed according to the Helsinki Declaration of 1975 and its amendments.

Data availability statement

The datasets generated and analyzed during the current study are not publicly available but can be obtained from the corresponding author upon reasonable request.

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Declaration of competing interest

The authors and funders declare no conflict of interest.

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Declaration of Generative AI and AI-assisted technologies in the writing process

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.iliver.2024.100142>.

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